Business Template

HOTEL DATABASE

Logo / Image

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1 BUSINESS DESCRIPTION

1.1 BUSINESS BACKGROUND

The hotel industry operates a complex ecosystem requiring seamless management of rooms, guests, bookings, staff, payments, and ancillary services. With multiple room types, varied check-in schedules, and diverse service offerings, coordinating this infrastructure efficiently presents a significant challenge. Hotel management needs a robust database system capable of tracking detailed information about rooms, reservations, customers, staff activities, room service orders, and payments. The goal is to maintain accurate real-time information on hotel operations and ensure consistently high service quality.

1.2 PROBLEMS. CURRENT SITUATION

Currently, hotel management faces several issues:

- Lack of a centralized system for tracking room availability, bookings, and customer data.
- Manual check-in and check-out processes, leading to errors and delays.
- Disjointed management of room service orders and staff assignments.
- Inability to quickly access guest history or generate accurate financial reports.

Without a unified database, managing these processes is inefficient, error-prone, and costly.

1.3 THE BENEFITS OF IMPLEMENTING A DATABASE. PROJECT VISION

The implementation of a robust database system would:

- Centralize all hotel-related information, improving coordination across departments.
- Automate booking workflows, room assignment, and check-in/check-out tracking.
- Provide real-time data on room status, occupancy, and service requests, improving the guest experience.
- Streamline payment processing, ensuring quick transactions and accurate financial records.

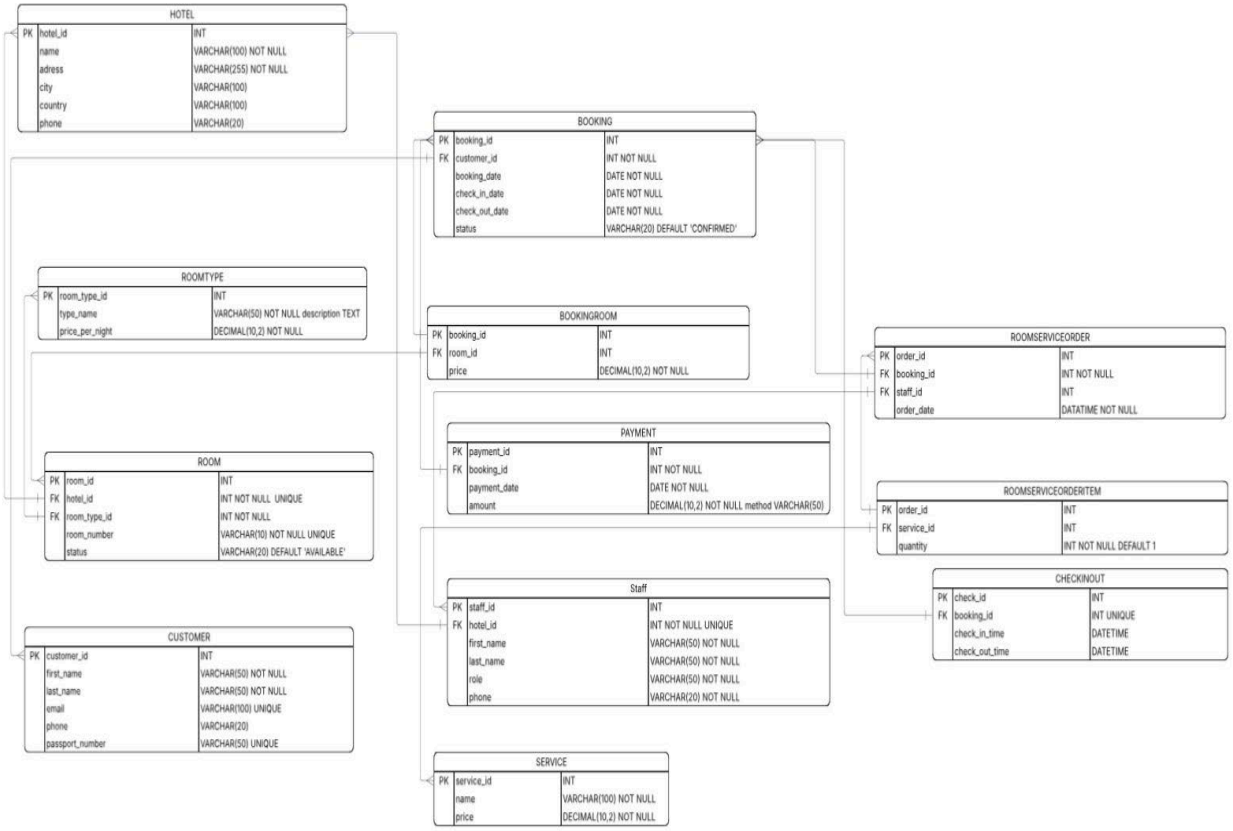
This database project aims to improve overall operational efficiency, reduce costs, and deliver guests a reliable and polished hospitality experience.

2 MODEL DESCRIPTION

2.1 DEFINITIONS & ACRONYMS

PK – Primary Key, UNIQUE identifier for each record in a table.
FK – Foreign Key, a reference to a primary key in another table.
Hotel – A lodging establishment offering rooms and services to guests.
Room – A rentable unit within a hotel, classified by type.
Booking – A reservation made by a customer for one or more rooms.
Customer – A person who makes a reservation and stays at the hotel.
Staff – An employee of the hotel responsible for a given role.
Service – An additional offering (e.g., laundry, spa) billed to a booking.
Payment – A financial transaction associated with a booking.

2.2 LOGICAL SCHEME



2.3 OBJECTS

In any type of relationship, the first one will be a table that describes. (one to many, one for the described table)

Table Description

Hotel table stores core information about each hotel property, including its name, address, city, country, and contact phone number.

Table Name	Field name	Field Description	Data Type
Hotel	hotel_id	Identifier, PK	INT

	name	Hotel name, NOT NULL	VARCHAR(100)
	address	Street address, NOT NULL	VARCHAR(255)
	city	City of the hotel	VARCHAR(100)
	country	Country of the hotel	VARCHAR(100)
	phone	Contact phone number	VARCHAR(20)

Relationships

Room (one-to-many), Staff (one-to-many)

Example with data

hotel_id	name	address	city	country	phone
1	Grand Plaza	Abay 10	Almaty	Kazakhstan	+7 727 000 0001
2	City Inn	Seyfullina 5	Astana	Kazakhstan	+7 717 000 0002

Table Description

RoomType table defines categories of rooms available in the hotel, including a description and the nightly rate.

Table Name	Field name	Field Description	Data Type
RoomType	room_type_id	Identifier, PK	INT
	type_name	Name of room type, NOT NULL	VARCHAR(50)
	description	Description of the room type	TEXT
	price_per_night	Nightly rate, NOT NULL	DECIMAL(10,2)

Relationships

Room (one-to-many)

Example with data

room_type_id	type_name	price_per_night
1	Standard	50.00
2	Deluxe	120.00
3	Suite	250.00

Table Description

Room table contains details of each individual room, linking it to a hotel and a room type, with its room number and current availability status.

Table Name	Field name	Field Description	Data Type
Room	room_id	Identifier, PK	INT
	hotel_id	Reference to hotel, FK, NOT NULL, UNIQUE	INT
	room_type_id	Reference to room type, FK, NOT NULL	INT
	room_number	Room number, NOT NULL, UNIQUE	VARCHAR(10)
	status	Availability status, DEFAULT 'AVAILABLE'	VARCHAR(20)

Relationships

Business Template

Hotel (many-to-one), RoomType (many-to-one), BookingRoom (one-to-many)

Example with data

room_id	hotel_id	room_type_id	room_number	status
1	1	1	101	AVAILABLE
2	1	2	201	OCCUPIED
3	2	3	301	AVAILABLE

Table Description

Customer table stores personal and contact information for each guest, including a unique email and passport number for identification.

Table Name	Field name	Field Description	Data Type
Customer	customer_id	Identifier, PK	INT
	first_name	First name, NOT NULL	VARCHAR(50)
	last_name	Last name, NOT NULL	VARCHAR(50)
	email	Email address, UNIQUE	VARCHAR(100)
	phone	Phone number	VARCHAR(20)
	passport_number	Passport number, UNIQUE	VARCHAR(50)

Relationships

Booking (one-to-many)

Example with data

customer_id	first_name	last_name	email	passport_number
1	Dias	Yermekov	dias@example.com	KZ001122334
2	Aisha	Bekova	aisha@example.com	KZ004455667

Table Description

Booking table records every reservation made by a customer, capturing the booking date, planned check-in and check-out dates, and current status.

Table Name	Field name	Field Description	Data Type
Booking	booking_id	Identifier, PK	INT
	customer_id	Reference to customer, FK, NOT NULL	INT NOT NULL
	booking_date	Date the booking was created, NOT NULL	DATE NOT NULL
	check_in_date	Planned check-in date, NOT NULL	DATE NOT NULL
	check_out_date	Planned check-out date, NOT NULL	DATE NOT NULL
	status	Booking status, DEFAULT 'CONFIRMED'	VARCHAR(20)

Relationships

Customer (many-to-one), BookingRoom (one-to-many), Payment (one-to-many), RoomServiceOrder (one-to-many), Checkinout (one-to-one)

Example with data

booking_id	customer_id	booking_date	check_in_date	check_out_date	status

1	1	2024-11-01	2024-11-08	2024-11-10	CONFIRMED
2	2	2024-11-03	2024-11-09	2024-11-12	CONFIRMED

Table Description

BookingRoom is a junction table that links bookings to specific rooms, recording the price agreed for that room at the time of booking.

Table Name	Field name	Field Description	Data Type
BookingRoom	booking_id	Reference to booking, PK/FK	INT
	room_id	Reference to room, PK/FK	INT
	price	Price for the room in this booking, NOT NULL	DECIMAL(10,2)

Relationships

Booking (many-to-one), Room (many-to-one)

Example with data

booking_id	room_id	price
1	1	50.00
1	2	120.00
2	3	250.00

Table Description

Payment table records all financial transactions related to a booking, including the payment date, amount, and payment method used.

Table Name	Field name	Field Description	Data Type
Payment	payment_id	Identifier, PK	INT
	booking_id	Reference to booking, FK, NOT NULL	INT NOT NULL
	payment_date	Date of payment, NOT NULL	DATE NOT NULL
	amount	Amount paid, NOT NULL	DECIMAL(10,2)
	method	Payment method (e.g. Card, Cash)	VARCHAR(50)

Relationships

Booking (many-to-one)

Example with data

payment_id	booking_id	payment_date	amount	method
1	1	2024-11-08	170.00	Card
2	2	2024-11-09	250.00	Cash

Table Description

Staff table stores employee details for each hotel, including their name, role, and contact phone. Each staff member is linked to a specific hotel and must have a unique assignment within it.

Table Name	Field name	Field Description	Data Type
Staff	staff_id	Identifier, PK	INT

	hotel_id	Reference to hotel, FK, NOT NULL, UNIQUE	INT NOT NULL
	first_name	First name, NOT NULL	VARCHAR(50)
	last_name	Last name, NOT NULL	VARCHAR(50)
	role	Employee role, NOT NULL	VARCHAR(50)
	phone	Contact phone, NOT NULL	VARCHAR(20)

Relationships

Hotel (many-to-one), RoomServiceOrder (one-to-many)

Example with data

staff_id	hotel_id	first_name	last_name	role
1	1	Aziz	Abikenov	Manager
2	1	Maira	Aslanova	Receptionist
3	2	Lucas	Thomas	Housekeeping

Table Description

Service table defines the catalog of additional services offered by the hotel (e.g., laundry, spa, airport transfer), each with a fixed price.

Table Name	Field name	Field Description	Data Type
Service	service_id	Identifier, PK	INT
	name	Service name, NOT NULL	VARCHAR(100)
	price	Service price, NOT NULL	DECIMAL(10,2)

Relationships

RoomServiceOrderItem (one-to-many)

Example with data

service_id	name	price
1	Laundry	15.00
2	Spa Treatment	80.00
3	Airport Transfer	35.00

Table Description

RoomServiceOrder table records each service request submitted during a booking, capturing which staff member handled it and the date/time of the order.

Table Name	Field name	Field Description	Data Type
RoomServiceOrder	order_id	Identifier, PK	INT
	booking_id	Reference to booking, FK, NOT NULL	INT NOT NULL
	staff_id	Reference to staff member, FK	INT
	order_date	Date and time of order, NOT NULL	DATETIME NOT NULL

Relationships

Booking (many-to-one), Staff (many-to-one), RoomServiceOrderItem (one-to-many)

Example with data

order_id	booking_id	staff_id	order_date
1	1	2	2024-11-08 10:00
2	2	3	2024-11-09 14:30

Table Description

RoomServiceOrderItem is a junction table linking each room service order to specific services, along with the quantity ordered. Quantity defaults to 1 if not specified.

Table Name	Field name	Field Description	Data Type
RoomServiceOrderItem	order_id	Reference to order, PK/FK	INT
	service_id	Reference to service, PK/FK	INT
	quantity	Quantity ordered, NOT NULL, DEFAULT 1	INT

Relationships

RoomServiceOrder (many-to-one), Service (many-to-one)

Example with data

order_id	service_id	quantity
1	1	2
1	3	1
2	2	1

Table Description

CheckInOut table tracks the actual check-in and check-out timestamps for each booking, providing an audit record of guest arrivals and departures. Each booking has exactly one check-in/out record.

Table Name	Field name	Field Description	Data Type
CheckInOut	check_id	Identifier, PK	INT
	booking_id	Reference to booking, FK, UNIQUE	INT UNIQUE
	check_in_time	Actual check-in date and time	DATETIME
	check_out_time	Actual check-out date and time	DATETIME

Relationships

Booking (one-to-one)

Example with data

check_id	booking_id	check_in_time	check_out_time
1	1	2024-11-08 14:00	2024-11-10 11:00
2	2	2024-11-09 15:30	NULL